

Public Information Content of Stock Prices in China

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Abstract

Reliable public information is the hallmark and cornerstone of a well-functioning financial market. This study evaluates stock market development in China by estimating the public information content of its stock prices. We show that overnight information is a good proxy for price informativeness of public information. It declined in China during 1996-2014 but increased in its neighbouring markets. The level of public information embedded in stock prices varies with market conditions and corporate governance variables. State ownership, ownership concentration, and political connection have negative but declining impact, while foreign ownership and high-quality auditors had an increasingly positive impact.

Keywords: Chinese stock market; public information; overnight information; audit; ownership structure;

JEL codes: G14; G15; G20; G30;

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I. Introduction

Public information plays a critical role in financial markets. High quality public information reduces volatility, risk premium, and the cost of capital (Diamond and Verrecchia, 1991; Li, 2005; Ai, 2010; Chang and Hong, 2016). It broadens investor base (Bernile, Kogan, and Sulaeman, 2015), reduces investor behavioural biases (Zhang, 2006), and improves market liquidity (Tetlock, 2010; Han, Tan, and Yang, 2016). While the overall market efficiency depends on the balance between public and private information, high quality public information is a precondition for financial market development as comprehensively argued by Black (2000). In this study, we estimate the public information content of stock prices in China during 1996-2014. It is used as the metric for its development in comparison to other markets. Over the sample period, China's stock markets have become the second largest market in the world with the highest trading value; see Figure 1 Panels A and B. They have experienced a wide range of regulatory and institutional reforms. Have these reforms led to a greater role of public information in pricing risky assets? Does China have an efficient and robust equity market that rivals its neighbours and the best in the world?

The literature on the quality of public information in China is growing in size and depth. There have been persistent doubts on the reliability of China's macroeconomic data due to technical deficiencies or outright manipulations at different government levels. Many studies have shown inconsistencies in the official GDP growth. Rawski (2001) speculates that the actual GDP growth in 1997-2001 was "no more than one-third of official claims." Holz (2004) deconstructs China's GDP statistics and find they are not comparable from year to year. Holz (2008) identifies "erroneous" national statistics that "casts doubt on the professionalism and sincerity of China's statistical authority." While "there is no clear evidence of data falsification, ... one is left with an issue of trust" (Holz, 2014).

At company level, the evidence is mixed whether various reforms have improved the market's overall information environment. Kuo, Ning, and Song (2014) report that the

abolition of the state-held non-tradable shares in 2005-06 “has not improved firms’ quality of financial information.” Although China adopted the International Financial Reporting Standard (IFRS) in July 2007, DeFond, et al. (2014) suggest that “China’s weak institutional infrastructure impairs IFRS’s ability to improve financial reporting quality.” Piotroski and Wong (2012) provide a comprehensive review of disclosure quality in China and conclude that “China’s financial markets continue to be plagued by weak information systems and, as a result, low-quality financial information.” They point out that “[G]iven the concentrated ownership structures, weak legal systems, highly politicized institutional arrangements, rent-seeking behavior, and corruption that characterize many developing economies, the benefits from opacity frequently outweigh the market and contracting-based benefits of transparency.” On the other hand, studies reviewed by Xie, Chi, and Liao (2016) show that the reduced state ownership since 2005 has improved accounting practice and corporate governance among listed firms in China. Chen, et al. (2016) show that lower state ownership and greater foreign participation increased the demand for high quality information and led to greater analyst coverage and better analyst performance. Using current market valuations to forecast future earnings, Carpenter, Lu, and Whitelaw (2015) conclude that “China’s stock market has become as informative about future corporate profits as in the US ... China’s stock market appears to be aggregating diffuse information and generating useful signals for managers.”

A unique feature of the Chinese stock markets is the omnipresence of the Chinese government and the Communist Party. At both macro and company levels, studies have found that political interference leads to information opacity. Holz (2008) accuses China’s National Bureau of Statistics at times acts like “a political propaganda organ ... with scarce or no regard for the compilation of accurate statistics.” Studies show that firms with political connections are more likely to engage in earnings management (Yang, Chi, and Young, 2012), to hire small local auditors (Wang, Wong, and Xia, 2008), and to involve in audit collusion (Chan, Lin, and Wong, 2010). Piotroski, Wong, and Zhang (2015) document strong evidence of listed firms

suppressing bad news around national meetings of the communist party and promotions of high-level provincial politicians. Overall the evidence indicates that political connection and interference are detrimental to the information environment of China's stock markets; see also Hope, Li, Liu, & Wu (2017) and Hope, Yue & Zhong (2017).

This study makes three contributions to the current literature. First, we empirically measure the broad public information content reflected in stock prices, not just firm-specific public information as in most existing studies. Macroeconomic conditions have a strong impact on company values. Yet the credibility of macroeconomic data has been absent in most studies of the information environment of China's stock market. It is difficult to measure price impact of events such as government planning (e.g. infrastructure projects), policy changes (e.g. foreign participation), regulatory changes (e.g. disclosure of insider trading), etc. Yet it is their price impact that reflects the credibility of public information as well as social trust of related public and private institutions; Holz (2014) and Li, Wang, & Wang (2017). The current literature has largely focused on more measurable aspects of the information environment, i.e. the quality of financial reporting by listed firms. By estimating the overall public information content of stock prices, we provide a broader view of the information environment in China's stock market and a wider assessment of institutional development.

Our second contribution is to propose a set of measures for the public information content embedded in stock prices. It is inherently difficult to separate the price impact of public versus private information. Our measures for public information are based on the information content of overnight returns when the market is closed. In China, continuous trading takes place for only four hours each day: 9:30 – 11:30 and 13:00 – 15:00. Most of the information events take place during the 18-hour "overnight" non-trading period. Companies tend to make major announcements after market close. Evening news is a major source of macro and political information. Online social media is a significant channel for information percolation and is most active during the early evening hours. Morning news reports price

movements in overseas markets. These observations suggest that overnight returns have a significant, perhaps dominant, public information component. Several studies have provided empirical support for the high private information content in daytime returns and the high public information content in overnight returns. French and Roll (1986) present evidence that “private information is the principle factor behind high trading-time variances” on the New York Stock Exchange. Fleming, Kirby, and Ostdiek (2006) show that in commodity markets where weather information is critical, overnight volatility accounts for a greater portion of the overall volatility. Based on textual analysis of the Dow Jones Newswire, Boudoukh, et al. (2016) show that “approximately 20%-40% of the overnight volatility can be tied to identifiable news compared to only 5%-6% of the trading day volatility.” Overall the literature provides strong support for high public information content in overnight returns.

We use the information component of overnight returns as a proxy for daytime (therefore overall) public information impact. We show that overnight information has the same effect as daytime public information in two cases. First, since trading is not necessary for public information to impact price, greater daytime public information should weaken the well-known volume-volatility relationship (e.g. Gallant, Rossi, and Tauchen (1992) and Jones, Kaul, and Lipson (1994)). Second, greater daytime public information at both firm and macro levels should increase the comovement between stock prices and the overall market. Indeed we find both effects for overnight information. It reduces the daytime volatility sensitivity to trading volume, and is negatively correlated with the price delay to market-wide information as measured by Chordia and Swaminathan (2000) and Hou and Moskowitz (2005). Our evidence indicates that overnight information flow has a strong positive correlation with daytime public information flow.

Our third contribution is to provide a long-run evaluation of the information environment in China’s stock markets while controlling variations in market conditions. Most studies of China’s equity markets present evidence on the impact of a particular institutional

feature, e.g. state ownership or political interference, on disclosure quality or price synchronicity. There is limited evidence on how the impact of a particular feature changes over time. We present evidence on how the institutional features have impacted the public information content over time. Even though the public information content of stock prices remains very low in China, we show that the negative impact from state ownership, ownership concentration, and political connection has been diminishing since 2011, while the positive impact from quality auditing and foreign ownership has been increasing. Furthermore, many studies fail to recognize and control for the impact of market conditions on the variable of interest, e.g. price synchronicity. Carpenter, Lu, and Whitelaw (2015) point out that the high price synchronicity in China is largely due to its high volatility. We show that market conditions, e.g. return, volatility, and liquidity, have a significant impact on the public information content of share prices. The recent decline of the public information content is largely due to changes in market conditions, not worsening of corporate governance in China.

Our empirical measures for the public information content of stock prices are estimated using price data from 1996 to 2014. We have access to ownership and auditing data from 2006 to 2014. Our empirical findings are summarized below:

- (i) Despite the extraordinary growth, the public information content of stock prices actually declined from 1996 to 2014 in China. After adjusting for the overnight impact from the S&P 500 returns, the night information measures in China declined 27% over the sample period, while they increased in Hong Kong, Korea, and Taiwan by an average of 72%. The night information measures of Chinese stocks are less than half of the values of the S&P 100 stocks. The daytime price delay measures have increased in China over the sample period. The long-run pattern of public information content is very similar to the price informativeness measure of Carpenter, Lu, and Whitelaw (2015) despite of very different data and methodologies.

- (ii) Market conditions, e.g. return, volatility, and liquidity, have a strong impact on the overnight information and price delay measures. The public information content increases with negative returns and high volatility, reflecting heightened market uncertainty and greater sensitivity to information. Illiquidity reduces price response to public information. The impact of market conditions helps to explain the variations in the public information content beyond corporate governance.
- (iii) To assess the impact of ownership structure and corporate governance, we use a two-stage regression to address the potential endogeneity in foreign ownership and high-quality auditors. We find that state ownership, ownership concentration, and political connection have negative effects on the public information content of stock prices. Their negative effect declined during 2011-14. Foreign ownership and high-quality auditors are associated with greater public information content and their impact increased in 2011-14. The decline in the public information content in 2009-2014 is largely due to changes in market conditions, e.g. relatively low volatility. It is not attributed to the ownership structure and corporate governance variables.

The rest of the paper has the following structure. Section II explains our data sample and summary statistics. Section III defines our measures for overnight information and discusses their historical patterns. Section IV presents the price delay measures and their historical patterns. Section V examines the determinants of the public information content of stock prices. We conclude in Section VI.

II. Data and Summary Statistics

Our trade data are extracted from the Thomson Reuters Tick History (TRTH) database and include daily opening and closing price as well as trading volume of each stock. Our sample period is from 1 January 1996 to 31 December 2014. In the early years, Chinese stocks were subject to frequent trading halts due to abnormal price movements. These trading halts

can be up to one or two weeks. For each year, we select a sample of stocks on the SSE that are (1) traded on at least on 80% of trading days in a year; (2) not suspended over 10 trading days in a year; (3) not special treated or delisted in a year. We construct a portfolio of small stocks and a portfolio of large stocks in each year. Each of the size portfolios has 50 stocks before 2005 and 100 stocks from 2006. We also download the daily Shanghai Composite Index, together with market index values for Hong Kong, Korea, and Taiwan. Our ownership and corporate governance data are from RESSET database. The earliest data available are from 2005. We extract state ownership, foreign ownership, ownership by the top 10 shareholders, and the auditor for each firm-year.

Table 1 provides the summary statistics of our sample. Panel A reports the statistics of number of sample firms, market value, annual return and standard deviation. The number of stocks satisfying our selection criteria is about 30% of all listed firms on SSE before 2010. At the end of 2009, the SSE revised the rules triggering trading halt of a listed company. Since then, around 60% to 80% of the stocks are traded more than 200 trading days per year. The sample stocks' average market value is higher than the average of all stocks in the early years. As the number of sample firms increases, the average market value gets closer the market average. The sample stocks had large swings in returns, mirroring the overall market returns and volatility in Panel C of Figure 1. For our sample, the average stock-year return is 8.1%, with standard deviation of 41%.

The state or foreign ownerships are the fractions of shares outstanding held by state or foreign shareholders. Audit is defined as a dummy variable that equal to 1 if a firm chooses the big 4 accounting firms as its independent auditor in a firm-year. Top 10 is the sum of ownerships of the top 10 shareholders, reflecting the level of ownership concentration. The level of state ownership serves as an indication of state control of the company. The change in state ownership reflects the balance of different economic and political interests. We use the change in state ownership as a proxy for political connection. Let S_t be the fraction of

shares held by the state in a firm-year and $\Delta S_t = S_t - S_{t-1}$. For $S_t > 0$, political connection is defined as $PC_t = \frac{\exp(\Delta S_t) - \exp(-1)}{\exp(+1) - \exp(-1)}$. Since $\max(\Delta S_t) = 1$ and $\min(\Delta S_t) = -1$, $PC_t \in [0,1]$. For $S_t = 0$, we define $PC_t = 0$. Since the calculation of political connection needs data of two years and the ownership data starts from 2005, the statistic of corporate governance variables and our subsequent analyses are based on the period of 2006 to 2014.

Panel B reports the statistics of corporate governance variables. It shows the average state ownership is about 25% in 2006. This is relatively low compared to state ownerships reported in earlier studies. For example, Gul, Kim, and Qiu (2010) report the average state ownership of 31.6% over 1996 – 2003. In 2005, the China government removed trading restrictions on state-owned shares, allowing the state-owned shares to be traded in the market. Across all firm-years, the variations of state ownership are very large. The 5 percentile has only 1.1% state ownership while the 95 percentile has 57.5%. The average foreign ownership is small at 4.2%, slightly higher than the 3.8% reported by Gul, Kim, and Qiu (2010). The cross-sectional average increased from 2.4% to 5.6% over the sample period. Across all firm-years, the bottom 5 percentile has zero foreign ownership while the top 5 percentile has almost 17% foreign ownership. Ownership concentration, as measured by the top 10 ownership, decline from 2006 to 2012 but increased in 2013-14. Again the variations across all firm-years are much larger. Political connection is around 0.27. The variation in the average PC is relatively small, reflecting the small year-to-year changes in the average state ownership. Across all firm-years, firms in the 95 percentiles have PC value that is 36% ($= \frac{0.310 - 0.228}{0.228}$) higher than those in the 5 percentile. The number of firms using big-4 as auditors increased sharply in 2011. Only around 2.5% listed firm employ big 4 accounting firm as their auditor during 2006-10. This number jumped to 22.2% in 2011, with an average of 18% for 2011-14. Only firms in the top 5 percentile use big-4 auditors.

III. Overnight Returns and the public information content of stock prices

In this section, we define three measures for the information content in overnight returns. We show that these measures weaken the daytime volume-volatility relationship, therefore are positively correlated with daytime public information. They are estimated each year to capture the changes in the public information content of stock prices on the SSE. We compare the long-run changes in China with its neighbouring markets. Our measures have similar patterns with the price informativeness measure of Carpenter, Lu, and Whitelaw (2015) despite of very different data and methodologies.

Measuring Overnight Information Flow

(i) The overnight information share (IS_N)

Hasbrouck (1995) first proposed the information share to compare price discovery across parallel markets where trading of the same asset takes place simultaneously. Wang and Yang (2011) extend the Hasbrouck model to the case of non-overlapping time periods or markets. Daily stock return r_t can be decomposed into two components: $r_t = m_t + e_t$ where m_t is serially independent and reflects the arrival of public or private information, e_t is serially correlated and reflects the price effect from bid-ask bounces, noise and liquidity trading, etc. Daily return $r_t = r_t^N + r_t^D$ where r_t^N is the overnight (close-to-open) return and r_t^D is the daytime (open-to-close) return. Overnight information flow is $\text{var}(m_t^N)$ and daytime information flow is $\text{var}(m_t^D)$. Clearly $m_t = m_t^N + m_t^D$ and $\text{var}(m_t) = \text{var}(m_t^N) + \text{var}(m_t^D)$. The overnight information share is defined as $IS_N = \frac{\text{var}(m_t^N)}{\text{var}(m_t)}$. To estimate IS_N , the daily return vector, $R_t = [r_t^N, r_t^D]'$, is modelled as a structural VAR process:

$$(1) \quad B_0 R_t = a + \sum_{k=1}^K B_k R_{t-k} + \eta_t,$$

where $\eta_t = [\eta_t^N, \eta_t^D]'$ is the vector of structural shocks and $a = [a^N, a^D]'$ is the vector of intercepts. The shocks η_t^N and η_t^D are serially uncorrelated, reflecting the overnight and daytime changes in economic fundamentals. We assume $E(\eta_t) = 0$; $E(\eta_t \eta_{t-k}') = 0$ for $k \neq 0$; $E(\eta_t \eta_t') = I$, a 2×2

identity matrix. Defining a trading day t as night followed by daytime, B_0 is a lower triangular matrix: for each day t , r_t^N affects r_t^D but not vice versa. Daytime trading affects the overnight return through the lagged returns on the right hand of the SVAR model.

The corresponding reduced form of the structural VAR is given by $A(L)R_t = \alpha + \varepsilon_t$, where $A(L) = I - A_1L - \dots - A_KL^K$, $A_k = B_0^{-1}B_k$, $\alpha = B_0^{-1}a$, and $\varepsilon_t = B_0^{-1}\eta_t$. Using the Beveridge-Nelson decomposition, Wang and Yang (2011) show that the random-walk component of daily return $r_t (=r_t^N + r_t^D)$ is given by $h'\eta_t$ where $h' \equiv [h_N, h_D] = \iota' A(1)^{-1} B_0^{-1}$ and $\iota = [1, 1]'$. The parameters h_N and h_D reflect the impact of the shocks η_t^N and η_t^D on the changes in the efficient price. The overnight information share is

$$(2) \quad IS_N = \frac{h_N^2}{h_N^2 + h_D^2}$$

The computation of IS_N is straightforward: $A(1)$ in the reduced-form VAR is estimated by OLS and the B_0^{-1} matrix is the lower triangle Cholesky factor of the variance matrix of ε_t .

(ii) *The overnight weighted price contribution (WPC_N)*

WPC is a non-parametric method that calculates the weighted average return attributed to a market or time period. It was first proposed by Barclay and Warner (1993) to measure relative price movements associated with different transaction sizes. It was later used to measure price discovery during the pre-opening period (Cao, et al. 2000), across trading venues (Huang, 2002), during overnight trading (Barclay and Hendershott, 2003, 2008), during opening and closing call auctions (Ellul, et al. 2005), and across dealers in different geographic locations (Anand, et al. 2010). We compare overnight price changes against daytime price changes. Over a period of T days, the overnight WPC is defined as

$$(3) \quad WPC_N = \sum_{t=1}^T \frac{r_t^N}{r_t} \left(\frac{|r_t|}{\sum_{\tau=1}^T |r_\tau|} \right) = \sum_{t=1}^T \frac{r_t^N}{r_t} w_t$$

So WPC_N is the weighted average of $\frac{r_t^N}{r_t}$. The weight $w_t = \frac{|r_t|}{\sum_{\tau=1}^T |r_\tau|}$ deals with the case where r_t is near or equal to zero. Daytime WPC, WPC_D , is defined similarly with $WPC_N + WPC_D = 1$.

(iii) R_N^2 of the unbiasedness regression

When examining price discovery during the pre-opening period, Biais, Hillion, and Spatt (1999) test for the presence of information trading in the pre-opening period using the following “unbiasedness regression”:

$$(4) \quad r_t = \alpha + \beta r_t^N + \varepsilon_t$$

The estimated β and the root mean squared errors are used to gauge the information content of overnight returns. Barclay and Hendershott (2008) argue that the OLS estimator of β can be interpreted as the overnight information-to-noise ratio. However most subsequent studies use the regression R^2 as the measure for the information content of overnight returns; see Cushing and Madhavan (2000), Madhavan and Panchapagesan (2000), Corwin and Lipson (2000), Chakrabarty et al. (2011), and Wang (2014), among others.

Overnight Information on the Shanghai Stock Exchange

Panel A of Table 2 presents the annual estimates of the NI measures. The time trends over the 19-year sample are depicted in Figure 2. The three NI measures have very similar values and high correlations around 0.9. The average of overnight information shares is around 8 to 9%. They are relatively low in 2003 and high in 1996-97 and 2007-08. While the estimates fluctuate yearly, the long-run trend is either flat or downward. Panel A of Table 2 also reports the differences in mean between the last four years (2011 to 2014) and previous periods. Overnight information in 2011-14 is lower than in 1996-99 but is higher than in 2003-06. The evidence indicates that overnight information, therefore the public information content of stock prices, has not increased over the sample period.

Panel B of Table 2 reports the NI measures for large and small stocks, which are depicted in Panels B and C of Figure 2. Overall large stocks have slightly higher NI measures than small stocks. The NI measures of large stocks declined in the first half of the sample period. They doubled in 2007-08 before reversing to a mild downward trend. The NI

measures of large stocks in 2012-14 are higher than those in 2003-05 but lower than those in 2009-11. Small stocks exhibit a downward trend over the full sample period. The NI measures of small stocks in 2011-14 are significantly lower than several earlier periods.

Figure 3 depicts the cross-sectional 5 and 95 percentiles of the NI measures for each year during the sample period. Firms at the 95 percentiles are the “good” firms whose prices have high public information content. Firms at the 5 percentiles are the “bad” firms whose prices have low public information content. We see two distinct patterns before and after 2006. Before 2006, the good firms do not always move in the same direction with the bad firms. After 2006, with the exception of R^2 , the good and bad firms tend to move in the same direction. For example, the correlations of the changes in WPC between the good and bad firms jumped from -0.026 in 1996-2006 to 0.772 in 2007-14. It seems that from 2007, the public information content of stock prices is less affected by firm-specific factors such as ownership structure and corporate governance, and is more determined by macro factors or the overall market condition. There is evidence, e.g. IS_N and R_N^2 , that the public information content of the good firms increased from 2009 to 2014.

Does Overnight Information Reflect Daytime Public Information?

Earlier we argued heuristically that overnight returns have a significant, perhaps dominant, public information component. We now seek empirical evidence that these measures are indeed positively and significantly correlated with daytime public information flow. The evidence will further justify using these measures to evaluate the overall price informativeness of public information. There is theoretical and empirical evidence that private information is incorporated into price through trading but public information can impact price without trading; e.g. Andersen (1996) and Fleming and Remolona (1999). Taken together, public information reduces the sensitivity of volatility to trading volume: volume may be low when volatility is mostly driven by public information. If our night information

measures are good proxies for daytime public information, they should have the same effect as daytime public information: reducing the sensitivity of volatility to volume. Since our night information measures are estimated annually, we test the hypothesis that, *ceteris paribus*, stocks with high overnight information have low volatility sensitivity to volume. The following cross-sectional regression is estimated annually:

$$(5) \quad \ln(\sigma_{i,n}^D) = \alpha + (\beta_V + \beta_{NIV}NI_{i,n})\ln(V_{i,n}) + \beta_{NI}NI_{i,n} + \beta_Z Z_{i,n} + \varepsilon_{i,n}$$

where $\sigma_{i,n}^D$ is the daytime (open-to-close) return volatility of firm i in year n ; $NI_{i,n}$ is the night information measure IS_N , WPC_N , or R_N^2 ; $V_{i,n}$ is the (daytime) trading volume; $Z_{i,n}$ is a set of control variables. The volatility elasticity to volume is $\beta_V + \beta_{NIV}NI_{i,n}$. Past studies have shown $\beta_V > 0$ and highly significant. Our null hypothesis is $\beta_{NIV} < 0$, i.e. overnight information decreases the (daytime) volatility sensitivity to volume.

The control variables include the 1-year lagged volatility due to its persistence. We also include market value at the start of the year ($MV_{i,n}$) and the Amihud illiquidity ($IL_{i,n}$). The signed returns are included to capture the asymmetric return effects on volatility. Let $r_{i,n}$ be the annual return. The signed returns are defined as $r_{i,n}^+ = r_{i,n}$ if $r_{i,n} > 0$ and 0 otherwise; $r_{i,n}^- = r_{i,n}$ if $r_{i,n} < 0$ and 0 otherwise. As indicated in Panel C of Figure 1, China's stock market had an extraordinary surge in 2007 and a massive crash in 2008. All firms in our sample had positive returns in 2007 and negative returns in 2008. These two years are excluded, resulting 17 annual cross-sectional regressions. Statistical inference is based on the adjusted Fama–MacBeth (FM) standard error as $\frac{\text{StDev}(\beta)}{\sqrt{T}} \sqrt{\frac{1+\rho(1)}{1-\rho(1)}}$, where $\rho(1)$ is β 's first-order autocorrelation and T is the number of years (Fama and French, 2002; Chakravarty, Gulen, Mayhew, 2004). The FM approach does not require a constant sample of stocks over the 19 years, which is suitable for the sample selection procedure in Section II.

The estimation results for (5) are reported in Table 3 and provide strong support for the null of $\beta_{NIV} < 0$: night information indeed reduces the daytime volatility elasticity to

volume, having the same effect as one would expect from daytime public information. The result holds for all three NI measures. In 17 annual cross-sectional regressions, β_{NIV} is negative and significant between 7 to 9 times. It is never positive and significant. The average volatility elasticity for a firm is measured by $\lambda_i \equiv \hat{\beta}_V + \hat{\beta}_{NI\sigma} \times NI_i$. We quantify the impact of NI for 2014, the last year in our sample. For a firm at 5 percentile IS_N , $IS_N = 0.017$ and $\lambda_{i,5\%} = 0.059 - 0.024 \times 0.017 = 0.05896$. For a firm at 95 percentile IS_N , $IS_N = 0.202$ and $\lambda_{i,95\%} = 0.059 - 0.024 \times 0.202 = 0.05458$. For the good firm with high IS_N , the volatility elasticity to trading volume is lower by $\frac{0.05458}{0.05896} - 1 = -7.4\%$. Similarly for WPC_N and R_N^2 , the differences in volatility elasticity to volume are -5.6% and -9.2% respectively.

The estimated β_{NI} is negative and significant at 5% for IS_N , WPC_N , and R_N^2 . Therefore high overnight information is associated with low daytime volatility. By definition, a high NI indicates a low daytime information share therefore a negative β_{NI} . Based on our heuristic arguments and the negative β_{NIV} , a high NI also indicates high public information relative to the total information flow. Therefore an alternative interpretation of the negative β_{NI} is that, ceteris paribus (especially with respect to trading volume), public information helps resolve market uncertainty, therefore reduces volatility. While we cannot attribute the negative β_{NI} to the latter mechanism only, we take the negative β_{NI} as supporting evidence that our NI measures capture the overall public information. The low overnight information in China relative to other markets (see below) may help explain its high volatility and price synchronicity; e.g. Carpenter, Lu, and Whitelaw (2015).

Comparison with Other Markets

While the overnight information shares have been relatively flat in China over the sample period, they have increased in China's neighbouring markets. We compare the overnight WPCs of the Shanghai Composite Index, the Hang Seng Index of Hong Kong, the KOSPI index of Korea, and the TAIEX index of Taiwan. The overnight WPCs are depicted in

Figure 4 Panel A. The cross-market differences of WPC_N are summarized in the top panel of Table 4. Panel A of Figure 4 shows that the overnight WPC in Shanghai and HKT markets diverged between 1996 and 2006, rose together in 2007-08, and declined together in 2009-14. The top panel of Table 4 shows that the overnight WPC declined 72% in Shanghai but increased an average of 90% for HKT markets. WPC_N in Shanghai surged 287% in 2007-08, while HKT markets had a mild increase of 31%. WPC_N in all markets declined in 2008-14, with Shanghai dropping 45% from its peak in 2008.

Even though we are not interested in the price impact of a particular source of public information, our aim is to estimate the price informativeness of local public information as a measure for market development in China. Trading in London and New York may have a significant impact on the overnight information in China. A strong impact from overseas markets may bias our estimates for the local public information content and our assessment of market development. To assess the impact of overseas markets, we regress overnight and daytime returns in Asian markets on the lagged daily returns of the S&P 500 index.¹ The residuals are then used to compute the overnight WPC for Asian markets via equation (3). Panel B of Figure 4 depicts overnight WPC after adjusting for the impact of the U.S. market return. The overall pattern remains the same as in Panel A. The changes in the adjusted WPC are reported in the bottom panel of Table 4. The adjusted WPC_N is lower than the unadjusted value by more than 30%: about one-third of the overnight information in Asia is from the US market. The fall, rise, and fall pattern for Shanghai is the same as in the top panel, as the rise, rise, and fall pattern for HKT markets. From 2008 to 2014, the adjusted WPC_N declined 7.8% for HKT markets and 40% for Shanghai, compared to 25% and 45% respectively for the unadjusted WPC_N . Therefore, the decline of WPC_N in HKT markets is mainly due to reduced

¹ We do not use New York daytime S&P 500 returns because S&P 500 futures are traded 24 hours on Globex.

impact from S&P 500 returns. But the decline of WPC_N in Shanghai is mainly due to reduced impact from local public information.

Panel C of Figure 4 compares the WPC_N in Shanghai against the IS_N , WPC_N , and R_N^2 of S&P 100 stocks. The average IS_N , WPC_N , and R_N^2 of S&P 100 stocks are 0.185, 0.183, and 0.192 respectively. They are 2 to 2.4 times the average NI measures of Chinese stocks reported in Table 2. The NI measures of S&P 100 stocks had a mild decline in 1996-2006. The average NI measures dropped from 0.17 to 0.14. They increased during the global financial crisis in 2007-08 to an average around 0.21. They peaked in 2011 during the European debt crisis, averaging around 0.31. The rise of the NI measures of S&P 100 stocks with the two financial crises suggests that they capture the public information content of stock prices in the United States. The public information content of stock prices in China is far below those of the developed markets and those of its neighbouring markets.²

Comparison with Price Informativeness of Carpenter, Lu, and Whitelaw (2015)

As mentioned earlier, evidence on the long-run change in the public information content of stock prices is very limited. Based on a sample from 1994 to 2000, Sami and Zhou (2004) show that accounting variables such as book value per share and earnings per share have high explanatory power for price variations in 1996-97. This coincides with the high NI measures in Table 2 and Figure 2. Recently Carpenter, Lu, and Whitelaw (CLW, 2015) present a comprehensive study of the development of China's equity markets. In particular, they investigate the changes in the informativeness of stock prices over the sample period of 1996 to 2012. Let E_n be the annual earnings, A_n be the book asset value, and M_n be the market value of a firm in year n . The following model is estimated each year:

$$\frac{E_{n+1}}{A_n} = c_n + a_n \ln\left(\frac{M_n}{A_n}\right) + b_n \left(\frac{E_n}{A_n}\right) + \varepsilon_{n+1}$$

² The NI measures of S&P 100 stocks are lower than those in Hong Kong, Korea, and Taiwan. Our conjecture is that US financial institutions have greater analytical capacity. The content of public information is partially priced in through private research before its official release. An example is Evans and Lyons (2012) where they present evidence that currency order flows forecast future economic fundamentals.

Price informativeness in year n is measured by the estimated coefficient $\hat{a}_n$, the predicted variation $\hat{a}_n\{\text{var}[\ln(M_n/A_n)]\}^{1/2}$, or the marginal increase in R^2 from adding $\ln(M_n/A_n)$ to the regression. Comparing the results of the Chinese firms with those of US firms studied by Bai, Philippon, and Savov (2016), CLW conclude that they have similar price informativeness.

Figure 5 compares CLW's predicted variation (PV) and marginal R^2 with our estimates of the overnight WPC in Table 2 and Figure 2 Panel A. Although the year-to-year variations are not always the same, the long-run trend, represented by the reverse-J shape from 1997-98 to 2007-08, is remarkably similar. WPC and CLW's R^2 are the lowest in 2003. CLW's PV and R^2 first peaked in 1998 and then in 2007, while WPC peaked in 1997 and 2008. From 2008 onwards, CLW's PV and R^2 are relatively flat, and so is our WPC during 2009-14. These measures are based on very different data and models. The common trend suggests that they are complementary measures of stock price informativeness. CLW links the long-run pattern in price informativeness to four development stages of the Chinese market. We show in Section V that variations in price informativeness are associated with changing market conditions and corporate governance in China.

IV. Price Delay and the public information content of stock prices

As a robustness check to our findings in Section III, we examine an alternative set of measures for the public information content of stock prices based on the delayed price adjustments to market-wide information. If market-wide information is widely available and widely regarded as price relevant, it should have an immediate impact on stock prices. If Chinese investors, institutional and individual, have increased their capacity to process public information, the delay of price adjustments should have a downward trend over the 19-year sample period. Chordia and Swaminathan (2000) show that stocks with high price delay have lower market value, fewer analysts, and lower volatility. Hou and Moskowitz (2005) document that price delay varies with investor attention variables, e.g. number of analysts,

number of shareholders, institutional ownership, stock size, and advertising expense. Griffin, Kelly, and Nardari (2010) report that price delay is inversely related to stock size in both developed and emerging markets. Callen, Khan, and Lu (2013) show that price delay is inversely related to the firm's accounting quality, e.g. accrual quality, special items, recent loss frequency and earnings surprise. Overall the existing studies support the use of price delay as an inverse measure for the public information content of stock prices. The price delay measures can be applied to daytime (open-to-close) returns, complementing our first set of measures based on overnight returns. The strong negative correlations between these two sets of measures further lend support to the use of overnight information as a proxy for the price informativeness of public information. The empirical findings based on price delays provide direct evidence on the public information content during daytime trading hours and serve as a robustness check on the findings based on overnight returns. We estimate three price delay (PD) measures proposed by Chordia and Swaminathan (2000, CS hereafter) and Hou and Moskowitz (2005, HM hereafter).

The HM price delay measures PD1 and PD2

As in Section III, daily return r_t of a stock can be decomposed into $r_t = r_t^N + r_t^D$ where r_t^N is the close-to-open return and r_t^D is the open-to-close return. We modify the HM model to estimate how the daytime return r_t^D of a stock react to the contemporaneous and lagged market returns $r_{M,t}^D$ and $r_{M,t}^N$. In our setting, r_t^N precedes r_t^D . The following regression is estimated for each stock-year in the sample:

$$(6) \quad r_t^D = \alpha + \beta^D r_{M,t}^D + \beta^N r_{M,t}^N + \sum_{k=1}^5 \beta_k r_{M,t-k} + \varepsilon_t$$

Let R^2 be the coefficient of determination of the regression in (6) and R_0^2 be the coefficient of determination of the regression $r_t^D = \alpha + \beta^D r_{M,t}^D + \varepsilon_t$. Parameters β^N and β_k ($k > 0$) capture the delayed impact of market-wide information on the price of a stock. HM's first delay measure is defined as $PD1 = 1 - \frac{R_0^2}{R^2}$. It is similar to a F-test on the joint significance of the lagged

market returns. The higher PD1, the more stock return variation is captured by the lagged market returns. By definition PD1 is inversely related to R_0^2 which is the widely used price synchronicity measure. The second measure of HM is based on the coefficients at different lags: $PD2 = \frac{\beta^N + \sum_{k=1}^5 k\beta_k}{\beta^D + \beta^N + \sum_{k=1}^5 \beta_k}$. It gives greater weights to the coefficients of longer lags, e.g. $k = 4$ or 5. HM also proposes a third delay measure based on the significance of lagged coefficients. We find its values to be very similar to PD2 and conduct our analyses using PD1 and PD2.

The CS price delay measure PD3

Similarly we modify the CS price delay to isolate daytime price delay only. For each year, the following Dimson regression is estimated for each stock in the sample:

$$(7) \quad r_t^D = \alpha + \sum_{k=-5}^{-1} \beta_k r_{M,t-k} + \beta^D r_{M,t}^D + \beta^N r_{M,t}^N + \sum_{k=1}^5 \beta_k r_{M,t-k} + u_t$$

The return of the market portfolio $r_{M,t}$ represents the market-wide information. Let $x \equiv \frac{\beta^N + \sum_{k=1}^5 \beta_k}{\beta^D}$. The price delay is defined as the logistic transformation of x , $PD3 = \frac{1}{1+e^{-x}}$. If there is no delayed price reaction, $\beta^N + \sum_{k=1}^5 \beta_k = 0$ and $PD3 = 0.5$. When there is delayed price reaction, $\beta^N + \sum_{k=1}^5 \beta_k > 0$ and $1 > PD3 > 0.5$. When there is price overreaction and reversal, $\beta^N + \sum_{k=1}^5 \beta_k < 0$ and $0 < PD3 < 0.5$. If price does not react to market-wide information in the same session, i.e. $\beta^D = 0$, $x = \infty$ and $PD3 = 1$.

Price Delay on the Shanghai Stock Exchange

The annual estimates of price delay on the SSE are reported in Panel A of Table 5 and are depicted in Figure 6 (with PD1 multiplied by 10). The PD measures were relatively low in 1996-97 when the night information measures were relatively high (Figure 2). They had large annual swings but no strong trend in 1998-2006. The decline in the price delay measures in 2007-08 occurred with the rise in the night information measures. The sharp rise in price delay in 2009-14 occurred with a mild decline in the night information measures. Average PD1 for large stocks in China is 0.068, close to HW's delay-sorted portfolio 6 (0.074). Table V

of CS reports price delay from 0.3 to 0.7 depending on market capitalization. The average PD3 in China is 0.616. Panel A of Table 5 also reports the differences in mean between the last four years (2011 to 2014) and previous periods. Price delay in 2011-14 is higher than in 1996-99 and 2007-10. Consistent with our finding on overnight information, price delay has not decreased on SSE over the sample period and has sharply increased in 2013-14.

The correlations of the estimated PD and NI measures are reported in Panel B of Table 5. While the three price delay measures have very different values, their correlations are very high at around 0.90. As expected, the estimated PD and NI measures are negatively correlated: stocks with low price delay have high price content of public information. By construction, the PD measures are strongly negatively correlated with the price synchronicity measure R_0^2 . The NI measures are positively correlated with R_0^2 . The correlation coefficients range from 0.31 to 0.47, suggesting that the NI measures capture information beyond price synchronicity with the market index.

V. Determinants of the public information content of stock prices

This section explores factors affecting the public information content of stock prices in China. Because of data limitation on institutional features such as ownerships, we first use the full sample to examine how market conditions affect the public information content. We then estimate the impact of institutional features in China during 2006 – 2014, controlling for the impact of market conditions.

Market Conditions and the Public Information Content of Stock Prices

The public information content of stock prices was very high in 1996-97 and 2007-08. A comparison between the NI measures in Panel A of Figure 2 with market volatility in Panel C of Figure 1 indicates a positive correlation between them: both are very low in 2003 and are very high in 1996-97 and 2007-08. Indeed the annual market volatility has a strong positive correlation (0.660) with the average WPC and negative correlation (-0.431) with the average

PD1. We also find that market return has a negative correlation (-0.320) with WPC and positive correlation (0.342) with PD1. These correlations suggest that the public information content of stock prices varies with market conditions.

To examine the link between market conditions and the public information content of stock prices, we estimate the following Fama-MacBeth regression:

$$(8) \quad Y_{i,n} = \alpha + \beta_{-1}Y_{i,n-1} + \beta_r r_{i,n} + \beta_\sigma \sigma_{i,n} + \beta_v v_{i,n} + \beta_{IL} IL_{i,n} + \varepsilon_{i,n}$$

$Y_{i,n}$ is the logit transformation of a NI or PD measure of stock i in year n ; $r_{i,n}$, $\sigma_{i,n}$, $v_{i,n}$ and $IL_{i,n}$ are the annual return, volatility, trading value, and the Amihud illiquidity respectively. As before, statistical inference is based on the adjusted Fama–MacBeth standard error. To assess any causal effects from market conditions to the public information content, we need to purge any endogenous effect from the NI and PD measures to market condition variables. The NI measures capture the information embedded in overnight returns. To remove the impact of overnight returns, we calculate $\hat{r}_{i,t}$ as the estimated constant plus the residuals from the “unbiasedness regression” in equation (4) and use it as the instrument for daily return $r_{i,t}$. By construction it is uncorrelated with overnight returns and highly correlated with daily returns. The adjusted annual return and volatility of stock i in year y as $\tilde{r}_{i,n} \equiv \sum_{t=1}^{T_n} \hat{r}_{i,t}$ and the annual daytime volatility is defined as $\tilde{\sigma}_{i,n} \equiv \sqrt{T_n \text{var}(\hat{r}_{i,t})}$, where T_n is the number of trading days in year n . A similar process is used to remove the effect of overnight returns on trading value and illiquidity. Since the PD measures are based on daytime returns, we purge any effect from daytime returns on market condition variables using the above procedure. The market condition variables purged off the effect of the NI and PD variables are denoted as $\tilde{r}_{i,n}$, $\tilde{\sigma}_{i,n}$, $\tilde{v}_{i,n}$, and $\tilde{IL}_{i,n}$.

Table 6 reports the estimated coefficients of the regression in (8) for the NI and PD measures. Except for overnight IS, the other NI and PD measures show significant year-to-year persistence. Volatility coefficients are positive for the NI measures and negative for the PD measures; therefore volatility increases the public information content and reduces price

delay. High volatility is generally associated with high information arrival, greater market uncertainty, and greater price sensitivity to information. The t statistics are significant at 5% for every year of the 19-year sample for overnight IS and WPC, and for 17 of the 19 years for overnight R^2 . For the PD measures, the t statistics are significant at 5% for 14 to 19 years of the 19-year sample. For all six measures, volatility has the highest explanatory power as indicated by the partial R^2 s. The surge in volatility in 2007-08 (Panel C of Figure 1) boosted our measures for the public information content of stock prices in those years. The reversal of our measures in 2010-14 is likely due to the relatively low market volatility.

Return coefficients are negative for the NI measures and positive for the PD measures; therefore return has a negative impact on the public information content of stock prices. This is consistent with investors paying greater attention to negative news. Investors, especially retail investors, tend to “ride the tide” when return is positive, and adjust their investments when return is negative.

Trading value has a strong time-series correlation with volatility, reflecting their dependence on the latent information arrival (Andersen, 1996). It also has a strong cross-sectional correlation with market value, reflecting the visibility of the stock and the size of its shareholders. High trading intensity is associated with greater public information content in prices. The Amihud illiquidity reduces IS and WPC and increases PD2. Greater illiquidity implies higher cost of trading, which dampens timely reaction to new information. Lin, et al. (2014) show that US stocks with high price delay have high Amihud illiquidity. Overall these market state variables explain a significant portion of the variations in the NI and PD measures. Based on their average partial R^2 s, the pecking order of the impact on overnight information is volatility, illiquidity, trading volume, and return; the pecking order of the impact on price delay is volatility, return, trading volume, and illiquidity.

To assess the changing explanatory power of market state variables, we show in Figure 7 the annual regression R^2 and the partial R^2 of the market state variables over the 19-year

sample period. The dependent variable is WPC in Panel A and PD1 in Panel B. For overnight WPC, the R^2 and the partial R^2 of all market state variables were very high in 2007-08 when the Chinese market had huge swings and extremely high volatility (Panel C of Figure 1). There is no strong time trend in the explanatory power of return and volatility, apart from the surge in 2007-08. Nor is there any trend for trading value since 2002. However, the explanatory power of illiquidity increased from an average of 3.41% during 1996-2006 to an average of 6.8% during 2009-14. This is the main reason for the rise of the total R^2 from an average of 20.2% during 1996-2006 to an average of 23.7% during 2009-14. For daytime price delay PD1, the explanatory power of return has a downward trend over the sample period. Volatility has a downtrend from 2000 to 2010, followed by an uptrend. Trading value and liquidity had little effect until after 2002. Contrary to WPC, trading value has greater explanatory power than illiquidity for PD1. Its explanatory power increased sharply in 2014.

Corporate Governance and the Public Information Content of Stock Prices

As documented by many studies, institutional features in China, e.g. state and foreign ownership, ownership concentration, political connect, etc. have strong impact on disclosure quality and the information environment in financial markets. We differ from previous studies in three aspects: First, we measure the public information content of stock prices, not price synchronicity. Second, while many studies use cross-sectional or panel data analyses to determine the impact of ownership structure, we evaluate how their impact changed over time. Third, most studies do not control for the impact of market conditions on price informativeness. We estimate the impact of corporate governance variables over time, while controlling the impact of market conditions. As seen from Table 6, market condition has strong effect on the public information content of stock prices.

Ownership structure is an important determinant of corporate governance, and is measured by state ownership (S), foreign ownership (F), and top-10 ownership (T10). We use

the change in state ownership as a proxy for the firm's political connection: $PC_t = \frac{\exp(\Delta S_t) - \exp(-1)}{\exp(+1) - \exp(-1)}$. A decrease (increase) of state ownership is interpreted as reflecting weak (strong) political connection. Auditing quality is captured by a dummy variable (A) that equals to one if the auditor is one of the big-4 international accounting firms.

Table 7 reports the correlations between WPC, PD1, stock size, and corporate governance variables. We see state ownership, ownership concentration (T10), and political connection are negatively correlated with the public information content of stock prices. Foreign ownership and the auditor dummy are positively correlated with the public information content of stock prices. Large firms tend to have high foreign ownership and high quality auditors: $\text{cor}(F, MV) = 0.32$ and $\text{cor}(A, MV) = 0.30$. State ownership is negative correlated with foreign ownership (-0.44) and high-quality auditors (-0.51). Foreign ownership is positively correlated with high-quality auditors (0.46), but is negatively correlated with ownership concentration (-0.46) and political connection (-0.23).

Since most firms are listed through the government's privatization program, we believe that state ownership, ownership concentration (T10), and political connection are determined by factors other than our measures of the public information content, therefore do not suffer any endogeneity problem. Foreign ownership may not be exogenous, e.g. foreign investors may be attracted to firms with good accounting practice and high disclosure quality. Similarly firms with good accounting practice and high disclosure quality are more likely to be large and to benefit more from using an internationally reputable auditor. To address this potential endogeneity, we use a two-stage regression as in Gul, Kim, and Qiu (2010). In the first stage, a probit model is estimated for the auditing dummy A:

$$(9) \quad \Pr(A_{i,n}) = \alpha + \beta_{MV}MV_{i,n} + \beta_S \ln(\text{Sales}_{i,n}) + \beta_L \text{Loss}_{i,n} + \beta_{Lev} \text{Lev}_{i,n} + \beta_{CR} \text{CR}_{i,n} + e_{i,n}$$

where $\Pr(A)$ is ex post coded as one for big-4 clients, and zero otherwise; MV is the market value; Sales is sales value in million yuan; Loss is a dummy that equals to one for loss firms,

and zero otherwise; Lev is the leverage ratio; CR is the current ratio. In general, profit-making large firms with high sales value and good financial health (low leverage and high current ratio) are more likely to use high-quality auditors. These same firms are also more likely to attract foreign ownership. Therefore we use the same set of explanatory variables to run the first-stage regression for foreign ownership.

Table 8 reports the estimated coefficients of the first-stage regressions. As expected, large firms with high sales tend to use big-4 auditors and attract more foreign investors. Financial health matters more when selecting auditors: big-4 clients have less loss and lower leverage. Foreign investors appear to be not sensitive to annual profitability. High foreign ownership is associated with higher leverage, perhaps reflecting more readily available bank credit (e.g. Blalock, Gertler, & Levine, 2008). The strength of these relationships varies from year to year. For the audit dummy, the estimated relationship in (9) becomes stronger over time with more coefficients being significant and higher regression R^2 . No such trend is observed in foreign ownership.

The fitted values of $\Pr(A)$ and F from (9), denoted as $\tilde{A}$ and $\tilde{F}$, are used as instruments in the second-stage regression:

$$(10) \quad Y_{i,n} = \alpha + \beta_{-1}Y_{i,n-1} + \beta_r\tilde{r}_{i,n} + \beta_\sigma\tilde{\sigma}_{i,n} + \beta_vMV_{i,n} + \beta_{IL}\tilde{L}_{i,n} \\ + \beta_S S_{i,n} + \beta_F\tilde{F}_{i,n} + \beta_A\tilde{A}_{i,n} + \beta_{T10}T10_{i,n} + \beta_{PC}PC_{i,n} + \varepsilon_{i,n}$$

As in (8), $Y_{i,n}$ is the logit transformation of a NI or PD measure of stock i in year n . Variables with $\sim$ are adjusted for endogeneity from the NI or PD measures. Because trading value and market value are highly correlated, we opt to control for stock size, replacing trading value with market value.

The estimation results of (10) are reported in Table 9. Panel A reports the results for the NI measures and Panel B reports the results for the PD measures. After including corporate governance variables, the impact of market condition variables, i.e. return, volatility, and illiquidity, are qualitatively the same as in Table 6. The coefficients of volatility

are statistically stronger but numerically larger for the NI measures and smaller for the PD measures. The coefficients for illiquidity are the opposite, much smaller for the NI measures and slightly larger for the PD measures. State ownership reduces overnight information and increases price delay. This negative effect is significant at 1% for all six measures. Foreign ownership has a strong positive impact on the public information content of stock prices. Ownership concentration (T10) and political connection reduce the public information content. High-quality auditors enhance the public information content, even though the estimated coefficients have low statistical significance. While our measures for price informativeness are very different from previous studies, the effects from corporate governance variables are qualitatively similar. Partial R^2 s indicate the pecking order of importance is state ownership, the auditor dummy, ownership concentration, foreign ownership, and political connection. Panel C of Table 9 reports the Hansen's J statistics for the endogeneity of the auditor dummy and foreign ownership. The p values are larger than the conventional significance levels, supporting the validity of the instruments.

The annual partial R^2 s of the corporate governance variables depicted in Figure 8 provide new evidence on the changing impact of difference variables over time. A striking feature is the surge in the importance of the auditor dummy and the decline of importance of ownership variables in 2011. State ownership and ownership concentration were very influential in 2008-10 but become less important in 2011-14. Foreign ownership becomes more important for WPC from 2012 and for PD1 in 2014. There is a significant and sustained decline of the importance of political connection from 2010. Although the overall public information content has not improved, the influence of the negative factors, i.e. state ownership, ownership concentration, and political connection, has decline. The influence of the positive factors, particularly the auditor dummy, has substantially increased.

VI. Conclusion

Based on two sets of measures, this study evaluates the long-run changes in a key aspect of capital market development in China, the significance of public information in the pricing of common stocks. Although the two sets of measures are very different, one based on overnight returns and the other based on daytime stock and market returns, both indicate that the public information content of stock prices has not increased in China, particularly in recent years. Low volatility in recent years is the main reason for the low price informativeness of public information. The negative impact from state ownership, ownership concentration, and political connection declined in 2010-14. The positive impact from quality auditors and foreign ownership has increased.

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Table 1: Summary Statistics

Panel A reports sample statistics on the number of sample firms (Firms) and the percentage of the total listed firms (Firm%), the average market value in billion yuan (MV) and its ratio to the average MV of all listed firms (MVR), the average return (r) and standard deviation (σ). Panel B reports the ownership percentages of the state (State), foreign investors (Foreign), the percentage audited by the big four accounting firms (Audit), the top 10 shareholders (Top10), and a proxy for political connection (PC).

Panel A: Sample firms, market value, and return

Year	Firms	Firms%	MV	MVR	r(%)	σ (%)
1996	118	36%	992	1.04	50.2	50.5
1997	102	30%	1875	1.92	60.1	47.9
1998	103	27%	1166	1.25	5.8	38.5
1999	120	26%	1530	1.90	15.8	43.1
2000	131	27%	2519	1.09	33.8	37.6
2001	166	27%	1832	1.32	-25.9	29.4
2002	181	30%	816	0.76	-32.2	29.5
2003	209	28%	1018	0.97	-7.4	28.7
2004	224	30%	2345	1.12	-18.5	35.4
2005	197	26%	741	0.88	-9.5	38.1
2006	205	27%	3689	1.61	50.1	40.0
2007	209	27%	22449	1.48	112.7	60.1
2008	259	33%	12350	1.39	-116.9	68.3
2009	281	36%	6580	0.75	75.9	49.6
2010	653	82%	14956	1.04	5.4	40.3
2011	663	79%	13824	0.82	-44.3	35.4
2012	687	76%	10633	0.77	-12.2	33.3
2013	686	75%	6521	0.96	-5.4	34.3
2014	615	64%	9207	0.99	16.2	31.7
Ave	306	41%	6055	1.16	8.08	40.6

Panel B: Ownership, audit, and political connection

	State(%)	Foreign(%)	Top10(%)	PC	Audit(%)
2006	26.8	2.4	60.9	0.265	2.4
2007	25.7	3.7	56.6	0.278	2.3
2008	23.5	3.3	57.5	0.267	2.5
2009	20.8	3.8	54.6	0.270	2.5
2010	22.5	4.4	53.4	0.277	2.8
2011	23.0	4.3	54.3	0.267	22.2
2012	21.5	4.7	54.3	0.264	12.2
2013	22.4	5.8	55.9	0.271	16.3
2014	23.0	5.6	57.8	0.272	21.4
mean	23.2	4.2	56.2	0.270	9.4
5%tile	1.1	0.0	32.9	0.228	0
50%tile	24.5	3.1	56.1	0.269	0
95%tile	57.5	16.9	76.6	0.310	100

Table 2: Annual Estimates of Overnight Information

This table reports the annual cross-sectional averages of IS_N , WPC_N , and R_N^2 and the differences in mean relative to 2011-14. The asterisks *, **, and *** indicate statistical significance at 10%, 5%, and 1% respectively.

Panel A: All stocks

	IS_N	WPC_N	R_N^2
1996	0.125	0.137	0.119
1997	0.164	0.140	0.149
1998	0.090	0.089	0.080
1999	0.103	0.092	0.110
2000	0.079	0.074	0.079
2001	0.087	0.086	0.072
2002	0.085	0.075	0.053
2003	0.063	0.055	0.039
2004	0.077	0.072	0.059
2005	0.078	0.069	0.057
2006	0.064	0.064	0.059
2007	0.082	0.089	0.080
2008	0.116	0.114	0.095
2009	0.076	0.079	0.077
2010	0.077	0.077	0.070
2011	0.107	0.102	0.094
2012	0.080	0.082	0.067
2013	0.105	0.078	0.080
2014	0.093	0.075	0.078
Average	0.092	0.087	0.080
$AVE_{2011-14}-AVE_{1996-99}$	-0.024***	-0.030***	-0.035***
$AVE_{2011-14}-AVE_{2000-02}$	0.008	0.003	0.001
$AVE_{2011-14}-AVE_{2003-06}$	0.026***	0.019***	0.026***
$AVE_{2011-14}-AVE_{2007-10}$	0.008	-0.006	-0.001

Panel B: Large and small stocks

	IS_N	WPC_N	R_N^2
Large stocks			
1996-99	0.120	0.115	0.123
2000-02	0.083	0.076	0.069
2003-05	0.069	0.063	0.051
2006-08	0.107	0.109	0.102
2009-11	0.131	0.137	0.126
2012-14	0.109	0.112	0.122
Average	0.104	0.103	0.100
$AVE_{2012-14} - AVE_{1996-99}$	-0.011	-0.003	-0.001
$AVE_{2012-14} - AVE_{2003-05}$	0.04***	0.049***	0.071***
$AVE_{2012-14} - AVE_{2009-11}$	-0.022**	-0.025***	-0.004**
Small stocks			
1996-99	0.104	0.116	0.088
2000-02	0.100	0.106	0.075
2003-05	0.105	0.122	0.070
2006-08	0.101	0.105	0.085
2009-11	0.095	0.094	0.069
2012-14	0.081	0.067	0.051
Average	0.098	0.102	0.074
$AVE_{2012-14} - AVE_{1996-99}$	-0.023***	-0.048***	-0.037***
$AVE_{2012-14} - AVE_{2003-05}$	-0.024***	-0.055***	-0.019**
$AVE_{2012-14} - AVE_{2009-11}$	-0.015**	-0.026***	-0.018*

Table 3: Volatility, Volume, and Overnight Information

This table reports the results of the following Fama-MacBeth regression:

$$\ln(\sigma_{i,n}^D) = \alpha + (\beta_V + \beta_{NIV}NI_{i,n})\ln(V_{i,n}) + \beta_{NI}NI_{i,n} + \beta_{-1}\ln(\sigma_{i,n-1}^D) + \beta_{MV}MV_{i,n} + \beta_{IL}IL_{i,n} + \beta_r^+r_{i,n}^+ + \beta_r^-r_{i,n}^- + \varepsilon_{i,n}$$

where $\sigma_{i,n}^D$ is the standard deviation of daytime (open-to-close) returns of firm i in year n ; $NI_{i,n}$ is the night information measure IS_N , WPC_N , or R_N^2 ; $V_{i,n}$ is the annual trading volume (million shares); $MV_{i,n}$ is the market value (billion yuan) at the start of year n ; $IL_{i,n}$ is the Amihud illiquidity; $r_{i,n}$ is the annual return with $r_{i,n}^+ = r_{i,n}$ if $r_{i,n} > 0$ and 0 otherwise; $r_{i,n}^- = r_{i,n}$ if $r_{i,n} < 0$ and 0 otherwise. The t statistics are based on the modified Fama-MacBeth standard error. “ $yrs(t \leq -1.96)$ ” and “ $yrs(t \geq 1.96)$ ” are the number of years with cross-sectional t statistics less than -1.96 or greater than 1.96 respectively. The asterisks ***, **, and * indicate statistical significance at 1%, 5%, and 10% respectively.

	β_V	β_{NIV}	β_{NI}	β_{-1}	β_{MV}	β_{IL}	β_r^+	β_r^-	α	R^2
NI = IS_N										
Coeff	0.059***	-0.024**	-0.434**	0.486***	-0.081***	0.328***	0.007	0.440***	-1.35***	0.444
t stat	3.58	-2.48	-2.08	7.45	-2.88	4.46	0.05	4.87	-2.94	
$yrs(t \leq -1.96)$	0	8	6	0	11	0	3	0	9	
$yrs(t \geq 1.96)$	9	0	0	17	2	10	4	12	2	
NI = WPC_N										
Coeff	0.052***	-0.024**	-0.831**	0.482***	-0.090***	0.441***	-0.005	0.436***	-1.56**	0.460
t stat	3.01	-2.25	-2.54	7.56	-3.21	4.20	-0.03	5.00	-2.54	
$yrs(t \leq -1.96)$	0	7	7	0	11	0	3	0	9	
$yrs(t \geq 1.96)$	9	0	0	17	2	11	3	11	3	
NI = R_N^2										
Coeff	0.060***	-0.032***	-0.539**	0.482***	-0.089***	0.357***	0.006	0.416***	-1.76***	0.452
t stat	5.55	-3.03	-2.28	7.29	-3.23	4.22	0.04	5.44	-3.32	
$yrs(t \leq -1.96)$	0	9	6	0	12	0	2	0	10	
$yrs(t \geq 1.96)$	10	0	0	17	2	11	3	12	2	

Table 4: Comparison with Hong Kong, Korea, and Taiwan

%Diff(WPC_{t1} , WPC_{t2}) is the percentage difference in WPC between $t1$ and $t2$.

	SHComp	HK	Korea	Taiwan	HKT Ave
Unadjusted for S&P 500 return					
WPC_{AVE}	0.092	0.393	0.336	0.332	0.354
%Diff(WPC_{1996} , WPC_{2006})	-72%	96%	114%	66%	90%
%Diff(WPC_{2006} , WPC_{2008})	287%	14%	43%	43%	31%
%Diff(WPC_{2008} , WPC_{2014})	-45%	-24%	-19%	-33%	-25%
%Diff(WPC_{2014} , WPC_{1996})	-40%	69%	148%	58%	85%
Adjusted for S&P 500 return					
Adj WPC_{AVE}	0.062	0.268	0.215	0.227	0.237
%Diff(WPC_{AVE} , Adj WPC_{AVE})	-32%	-32%	-36%	-32%	-33%
%Diff(Adj WPC_{1996} , Adj WPC_{2006})	-63%	112%	23%	11%	45%
%Diff(Adj WPC_{2006} , Adj WPC_{2008})	233%	28%	15%	43%	29%
%Diff(Adj WPC_{2008} , Adj WPC_{2014})	-40%	-20%	20%	-13%	-7.8%
%Diff(Adj WPC_{2014} , Adj WPC_{1996})	-27%	116%	70%	37%	72%

Table 5: Annual Estimates of Price Delay

This table reports the annual cross-sectional averages of PD1, PD2, and PD3 and the differences in mean relative to 2011-14. The asterisks *, **, and *** indicate statistical significance at 10%, 5%, and 1% respectively.

Panel A: All stocks

	PD1	PD2	PD3
1996	0.027	0.450	0.576
1997	0.032	0.509	0.584
1998	0.092	0.722	0.649
1999	0.082	0.706	0.630
2000	0.139	0.867	0.657
2001	0.075	0.581	0.600
2002	0.066	0.527	0.595
2003	0.073	0.775	0.629
2004	0.069	0.689	0.607
2005	0.045	0.608	0.608
2006	0.085	0.746	0.645
2007	0.037	0.510	0.595
2008	0.027	0.401	0.573
2009	0.043	0.565	0.615
2010	0.054	0.586	0.621
2011	0.049	0.552	0.606
2012	0.047	0.573	0.598
2013	0.086	0.720	0.631
2014	0.173	0.887	0.684
Average	0.068	0.630	0.616
AVE ₂₀₁₁₋₁₄ -AVE ₁₉₉₆₋₉₉	0.030***	0.086***	0.020***
AVE ₂₀₁₁₋₁₄ -AVE ₂₀₀₀₋₀₂	-0.004	0.025	0.013
AVE ₂₀₁₁₋₁₄ -AVE ₂₀₀₃₋₀₆	0.021	-0.022	0.007
AVE ₂₀₁₁₋₁₄ -AVE ₂₀₀₇₋₁₀	0.049***	0.167***	0.029***

Panel B: Correlations of the NI and PD measures

	IS _N	WPC _N	R _N ²	PD1	PD2	PD3
WPC _N	0.92					
R _N ²	0.87	0.89				
PD1	-0.61	-0.51	-0.50			
PD2	-0.58	-0.51	-0.45	0.89		
PD3	-0.62	-0.53	-0.47	0.89	0.94	
R ₀ ²	0.44	0.47	0.31	-0.70	-0.82	-0.88

Table 6: Market Condition and Price informativeness of Public Information

This table reports the results of the Fama-MacBeth estimation of the following regression:

$$Y_{i,n} = \alpha + \beta_{-1}Y_{i,n-1} + \beta_r\tilde{r}_{i,n} + \beta_\sigma\tilde{\sigma}_{i,n} + \beta_{TV}\tilde{TV}_{i,n} + \beta_{IL}\tilde{IL}_{i,n} + \varepsilon_{i,n}$$

$Y_{i,n}$ is the logit transformation of a NI or PD measure of stock i in year n ; $\tilde{r}_{i,n}$, $\tilde{\sigma}_{i,n}$, $\tilde{TV}_{i,n}$ and $\tilde{IL}_{i,n}$ are the return, volatility, trading value, and illiquidity respectively, adjusted for endogeneity. We also report the number of years (out of 19) with $t \leq -1.96$ and $t \geq 1.96$ and the partial R^2 of the market state variables.

$Y_{i,n}$	β_{-1}	β_r	β_σ	β_{TV}	β_{IL}	α	$\bar{R}^2$
IS	0.037	-0.126**	0.259***	0.042	-0.819***	-2.76***	0.231
<i>t stat</i>	1.20	-2.34	2.68	0.94	-4.01	-2.60	
<i>yrs(t ≤ -1.96)</i>	0	12	0	0	8	14	
<i>yrs(t ≥ 1.96)</i>	3	1	19	4	0	0	
PR ²	0.008	0.021	0.149	0.009	0.041		
WPC	0.083***	-0.079***	0.302**	0.088***	-1.21***	-3.67**	0.251
<i>t stat</i>	3.79	-2.76	2.40	3.66	-3.04	-2.21	
<i>yrs(t ≤ -1.96)</i>	0	7	0	5	14	17	
<i>yrs(t ≥ 1.96)</i>	9	0	19	7	0	0	
PR ²	0.015	0.018	0.121	0.027	0.056		
R_N^2	0.084***	-0.226***	0.137**	0.088*	-0.135	-4.05***	0.186
<i>t stat</i>	2.88	-3.31	2.00	1.71	-0.52	-5.07	
<i>yrs(t ≤ -1.96)</i>	0	13	0	1	2	18	
<i>yrs(t ≥ 1.96)</i>	6	0	17	9	1	0	
PR ²	0.009	0.026	0.117	0.019	0.005		
PD1	0.139***	0.442***	-0.442***	-0.069***	0.205	-1.79***	0.169
<i>t stat</i>	4.49	3.21	-4.26	-3.81	1.23	-4.93	
<i>yrs(t ≤ -1.96)</i>	0	0	16	6	1	10	
<i>yrs(t ≥ 1.96)</i>	11	10	0	0	5	2	
PR ²	0.028	0.024	0.060	0.016	0.006		
PD2	0.069***	0.343***	-0.456***	-0.043*	0.222**	1.02**	0.102
<i>t stat</i>	4.09	3.04	-5.34	-1.68	2.19	1.96	
<i>yrs(t ≤ -1.96)</i>	0	0	14	4	1	2	
<i>yrs(t ≥ 1.96)</i>	6	10	0	2	6	8	
PR ²	0.011	0.017	0.057	0.011	0.009		
PD3	0.127***	0.127***	-0.139***	-0.012***	0.009	0.487***	0.212
<i>t stat</i>	3.49	4.89	-3.85	-3.17	0.17	4.85	
<i>yrs(t ≤ -1.96)</i>	0	0	19	6	1	0	
<i>yrs(t ≥ 1.96)</i>	13	14	0	0	4	10	
PR ²	0.025	0.039	0.113	0.010	0.002		

Table 7: Correlations of Public Information Measures and Firm Characteristics

This table reports the cross-sectional average correlations of price informativeness measures (WPC_N and PD1) and firm characteristics.

	WPC	PD1	MV	Illiq	Return	St Dev	State	Foreign	Audit	T10
PD1	-0.62									
MV	0.19	-0.15								
Illiq	-0.01	-0.04	-0.56							
Return	-0.23	0.17	-0.06	-0.04						
St Dev	0.27	-0.23	-0.14	0.29	0.06					
State	-0.55	0.55	-0.20	0.10	0.04	-0.10				
Foreign	0.38	-0.41	0.32	-0.15	0.03	0.06	-0.44			
Audit	0.45	-0.35	0.30	-0.31	0.02	0.11	-0.51	0.46		
T10	-0.52	0.52	-0.11	0.08	0.04	-0.09	0.61	-0.46	-0.52	
PC	-0.33	0.31	-0.12	0.06	0.09	-0.06	0.37	-0.23	-0.28	0.34

Table 8: Instrumental Variables

This table reports the estimation of the following regression:

$$Y_{i,n} = \alpha + \beta_{MV}MV_{i,n} + \beta_S \ln(\text{Sales}_{i,n}) + \beta_L \text{Loss}_{i,n} + \beta_{Lev} \text{Lev}_{i,n} + \beta_{CR} \text{CR}_{i,n} + e_{i,n}$$

where Y is either the audit dummy A or foreign ownership F; MV is the market value; Sales is sales value in million yuan; Loss is a dummy that equals to one for loss firms, and zero otherwise; Lev is the leverage ratio; CR is the current ratio.

	β_{MV}	β_S	β_L	β_{Lev}	β_{CR}	α	R^2
Audit	0.022***	0.074***	-0.014*	-0.051***	0.042***	-1.433***	0.286
<i>t-value</i>	2.75	17.8	-1.81	-5.04	4.61	-10.9	
<i> yrs(t<=-1.96)</i>	0	0	1	8	0	7	
<i> yrs(t>=1.96)</i>	2	9	0	0	8	0	
Foreign	0.653**	1.526***	-0.531	0.075***	0.817***	-0.147***	0.237
<i>t-value</i>	2.44	7.71	-1.33	3.17	3.35	-2.70	
<i> yrs(t<=-1.96)</i>	0	0	1	0	0	2	
<i> yrs(t>=1.96)</i>	1	8	0	2	1	0	

Table 9: Corporate Governance and Price informativeness of Public Information

This table reports the results of the Fama-MacBeth estimation of the following regression:

$$Y_{i,t} = \alpha + \beta_{-1}Y_{i,t-1} + \beta_r \tilde{r}_{i,t} + \beta_\sigma \tilde{\sigma}_{i,t} + \beta_{MV} MV_{i,t} + \beta_{IL} \tilde{L}_{i,t} + \beta_S S_{i,t} + \beta_F \tilde{F}_{i,t} + \beta_A \tilde{A}_{i,t} + \beta_{T10} T10_{i,t} + \beta_{PC} PC_{i,t} + \varepsilon_{i,t}$$

$Y_{i,t}$ is the logit transformation of a NI or PD measure of stock i in year t ; $r_{i,t}$, $\sigma_{i,t}$, and $L_{i,t}$ are the return, volatility, and illiquidity respectively; MV is the firm's market value at the start of the year; S and F are the state and foreign ownerships, respectively, at the start of the year; $A = 1$ if the auditor is one of the big four and 0 otherwise; PC is a proxy for political connection. The t statistics are based on the adjusted Fama-MacBeth standard error.

Panel A: The NI measures

	β_{-1}	β_r	β_σ	β_{MV}	β_{IL}	β_S	β_F	β_A	β_{T10}	β_{PC}	α	R^2
IS	0.063*	-0.294*	0.322***	0.076**	-0.065**	-0.721***	1.925***	0.170*	-0.273*	-0.082***	-3.38***	0.404
<i>t stat</i>	1.81	-1.86	5.95	2.07	-2.56	-5.21	3.60	1.69	-1.75	-2.65	-5.40	
<i>yrs(t <= -1.96)</i>	0	6	0	0	3	5	0	0	3	2	6	
<i>yrs(t >= 1.96)</i>	4	1	9	3	0	0	4	3	0	0	0	
PR ²	0.014	0.006	0.105	0.009	0.008	0.045	0.030	0.024	0.005	0.014		
WPC	0.076***	-0.130***	0.241***	0.055***	-0.039**	-0.747***	2.94***	0.224	-0.788***	-0.100**	-1.79***	0.531
<i>t stat</i>	2.90	-3.23	5.67	2.61	-2.06	-5.72	6.88	1.61	-4.95	-2.54	-4.39	
<i>yrs(t <= -1.96)</i>	0	4	0	1	2	7	0	0	6	2	7	
<i>yrs(t >= 1.96)</i>	5	0	9	5	0	0	6	4	0	0	0	
PR ²	0.022	0.011	0.117	0.023	0.006	0.067	0.029	0.034	0.045	0.015		
R_N^2	0.052*	-0.394**	0.218***	0.168***	-0.021**	-1.042***	2.297***	0.195*	-0.792**	-0.079**	-2.99***	0.324
<i>t stat</i>	1.87	-2.26	6.69	5.52	-2.05	-5.26	2.65	1.78	-2.56	-2.02	-4.58	
<i>yrs(t <= -1.96)</i>	0	5	0	0	2	6	0	0	3	1	7	
<i>yrs(t >= 1.96)</i>	2	0	8	6	0	0	3	3	0	0	0	
PR ²	0.007	0.012	0.082	0.031	0.012	0.040	0.018	0.020	0.014	0.013		

Panel B: The PD measures

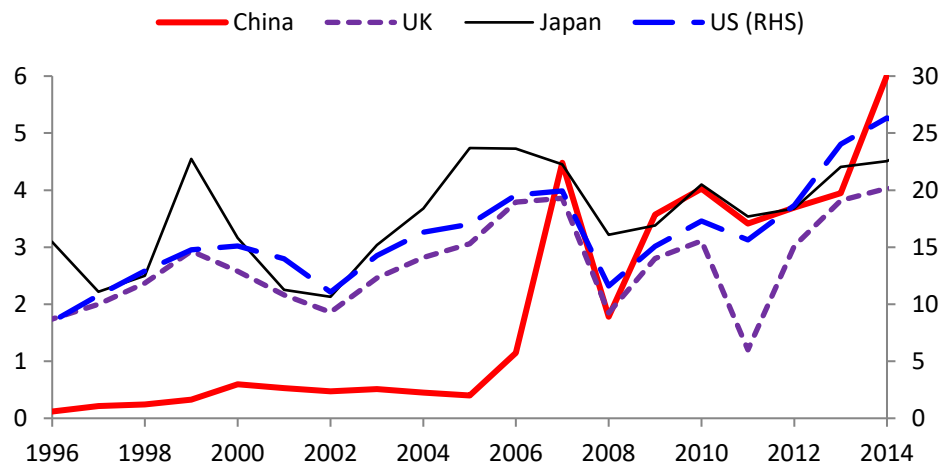
	β_{-1}	β_r	β_σ	β_{MV}	β_{IL}	β_S	β_F	β_A	β_{T10}	β_{PC}	α	R^2
PD1	0.134**	0.374***	-0.248***	-0.017**	0.251***	1.63***	-3.98***	-0.299**	1.28***	0.136***	-2.92***	0.401
<i>t stat</i>	2.40	2.95	-11.6	-2.57	2.44	5.96	-2.65	-1.99	4.72	2.90	-4.98	
<i>yrs(t<=-1.96)</i>	0	0	9	4	0	0	4	4	0	0	6	
<i>yrs(t>=1.96)</i>	3	3	0	0	2	7	0	1	6	2	0	
PR ²	0.005	0.005	0.068	0.011	0.003	0.045	0.022	0.049	0.030	0.006		
PD2	0.032*	0.289***	-0.230***	-0.040***	0.319***	1.14***	-3.98***	-0.319*	1.18***	0.080**	-2.61**	0.340
<i>t stat</i>	1.81	3.51	-6.11	-2.81	4.28	3.66	-2.80	-1.87	3.92	2.31	-2.54	
<i>yrs(t<=-1.96)</i>	0	0	7	4	0	0	3	3	0	0	4	
<i>yrs(t>=1.96)</i>	2	3	0	1	4	6	0	0	5	3	1	
PR ²	0.005	0.013	0.086	0.012	0.019	0.030	0.012	0.018	0.020	0.004		
PD3	0.095***	0.271***	-0.178***	-0.015	0.102*	0.127***	-1.24***	-0.114**	0.076	0.054**	-0.267***	0.265
<i>t stat</i>	3.10	3.05	-6.72	-1.58	1.72	4.25	-2.88	-2.51	1.57	2.10	-3.92	
<i>yrs(t<=-1.96)</i>	0	0	9	3	1	0	3	3	0	0	6	
<i>yrs(t>=1.96)</i>	5	6	0	0	3	4	0	0	2	3	0	
PR ²	0.018	0.015	0.069	0.016	0.005	0.019	0.008	0.010	0.004	0.003		

Panel C: Hansen's J test

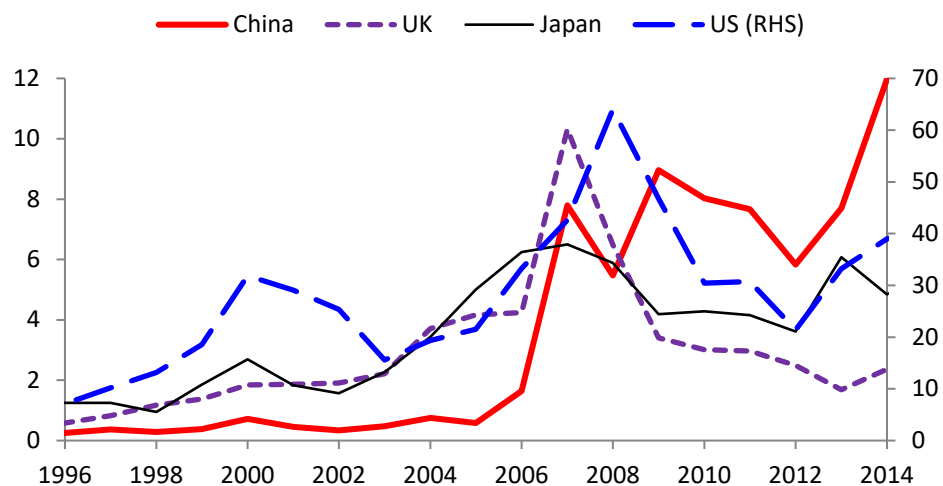
	WPC	IS	R_N^2	PD1	PD2	PD3
A	3.296	3.228	2.769	3.451	4.745	4.551
P-value	0.51	0.52	0.60	0.49	0.31	0.34
F	1.467	1.498	1.377	2.520	2.410	2.085
P-value	0.83	0.83	0.85	0.64	0.66	0.72

Figure 1: China's Equity Market

Panel A: Market capitalization (trillion USD)



Panel B: Trading value (trillion USD)



Panel C: Annual return and volatility of the Shanghai Composite Index

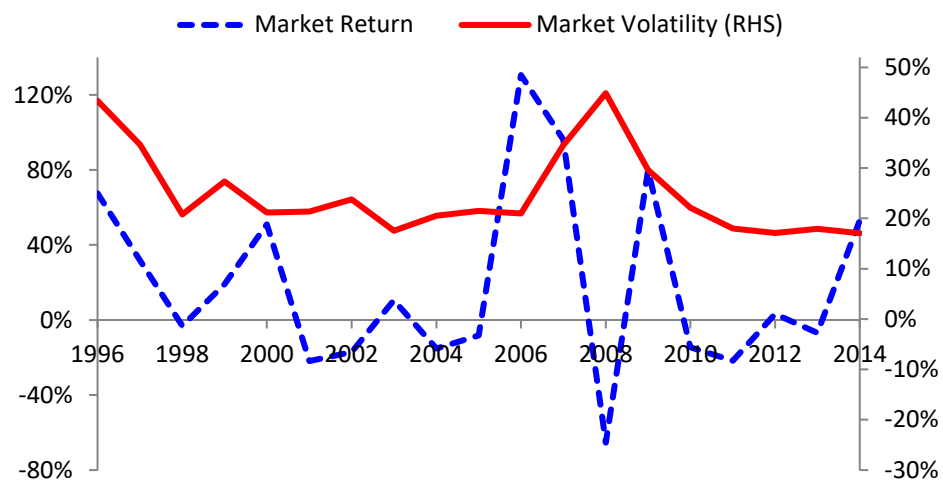
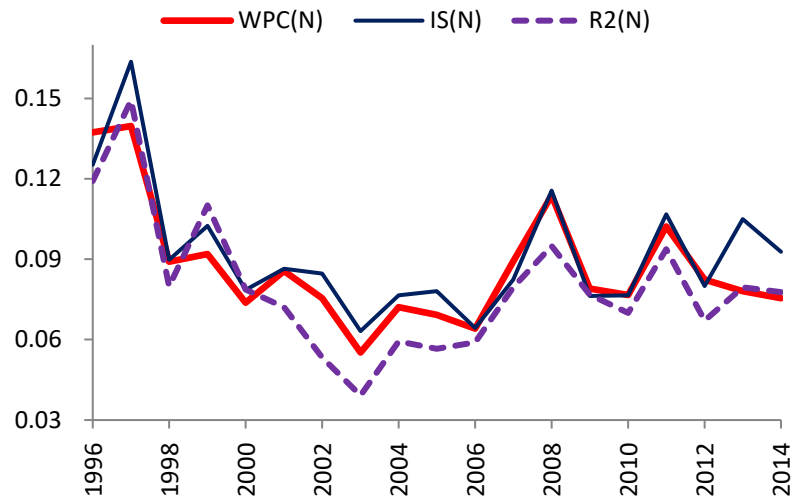


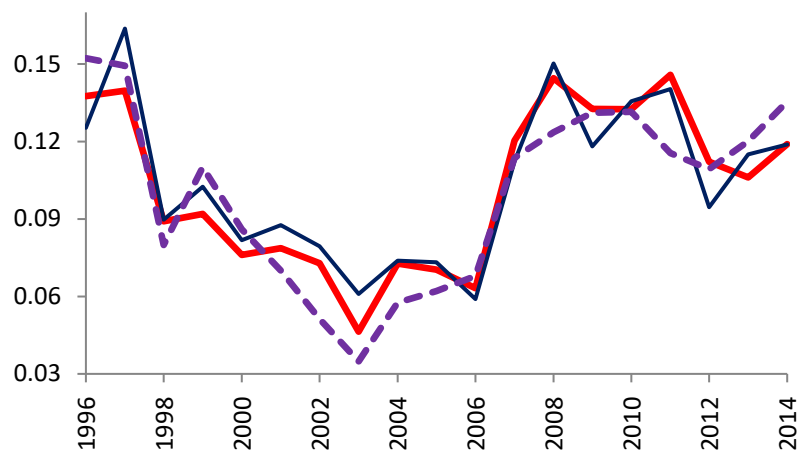
Figure 2: Annual Estimates of Overnight Information

This figure depicts the annual estimates of the overnight information share IS_N , overnight WPC_N , and R_N^2 of the unbiasedness regression.

Panel A: All stocks



Panel B: Large stocks



Panel C: Small stocks

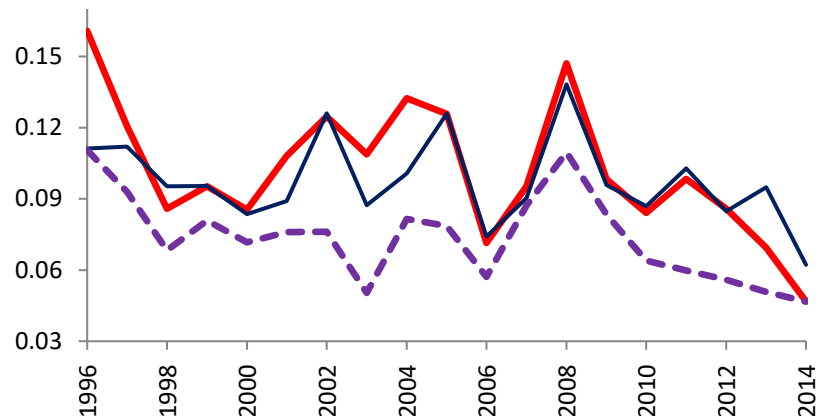
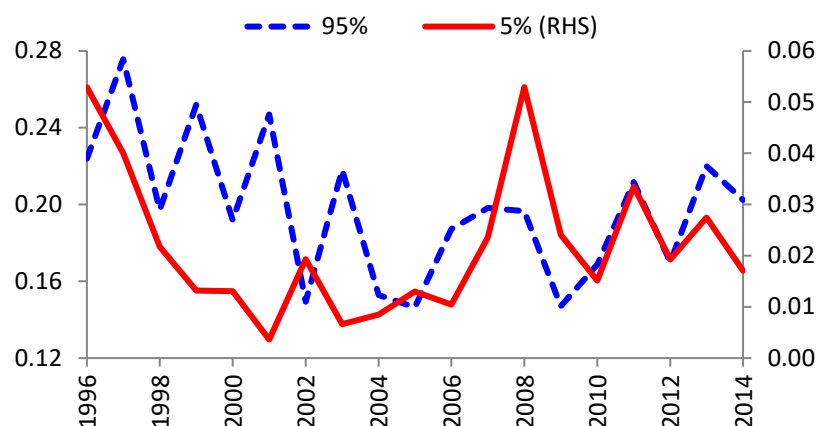


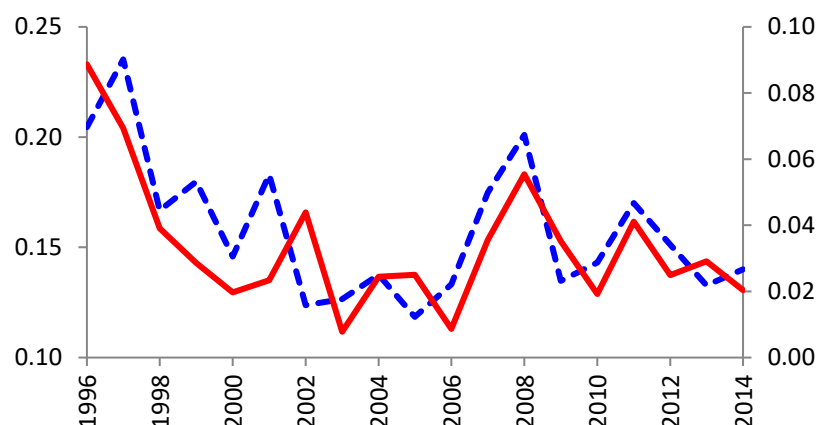
Figure 3: Firms with High and Low Public Information Content

This figure depicts the 95 and 5 percentiles of the estimated overnight information share IS_N , overnight WPC_N , and R_N^2 of the unbiasedness regression. The red solid line (5%) represents firms with low overnight information. The blue dashed line (95%) represents firms with high overnight information.

Panel A: Overnight information share IS_N



Panel B: overnight WPC_N



Panel C: Overnight R_N^2

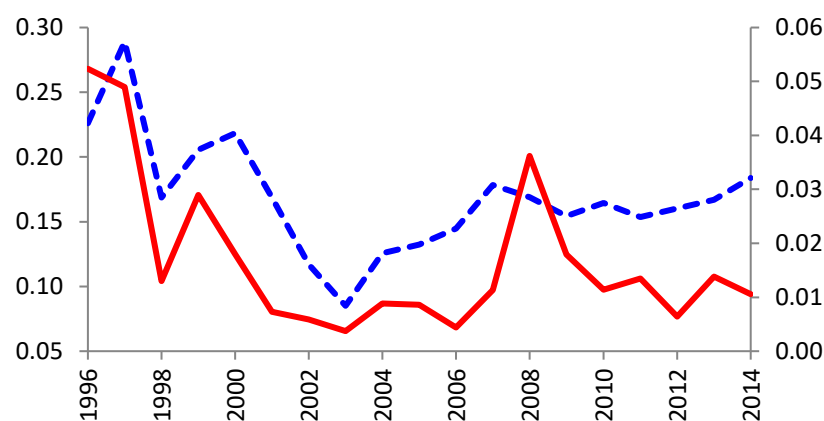
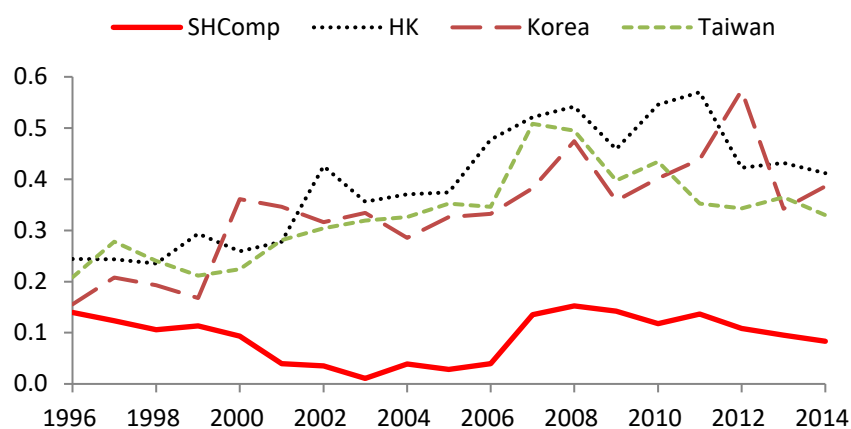
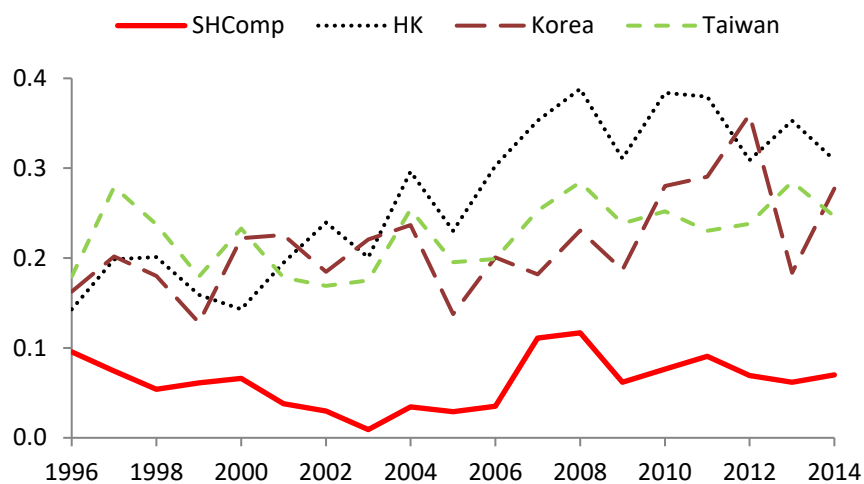


Figure 4: Overnight Information Shares in Asia and U.S.

Panel A: WPC_N based on raw overnight returns



Panel B: WPC_N based on overnight returns adjusted for the S&P 500 index returns



Panel C: WPC_N of Chinese stocks and IS_N , WPC_N , and R_N^2 of S&P 100 stocks

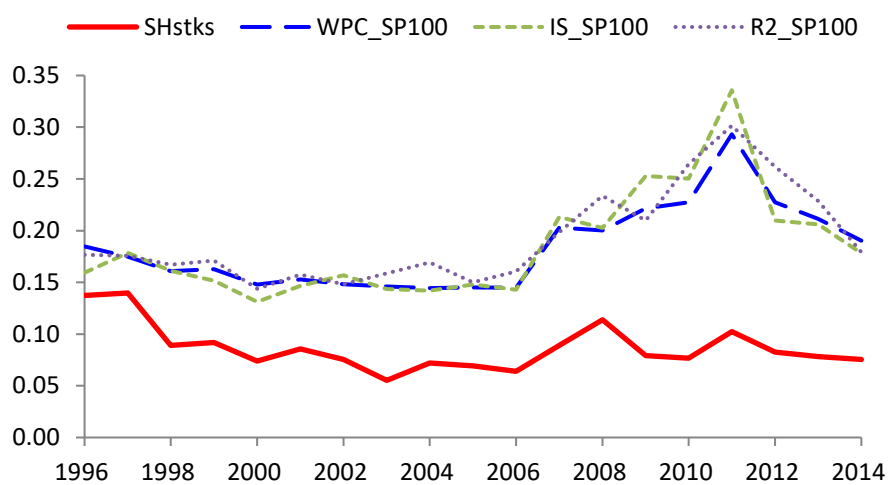


Figure 5: Comparison with CLW (2015)

This figure compares the overnight WPC with the public information content measures CLW R2 and CLW PV of CLW (2015).

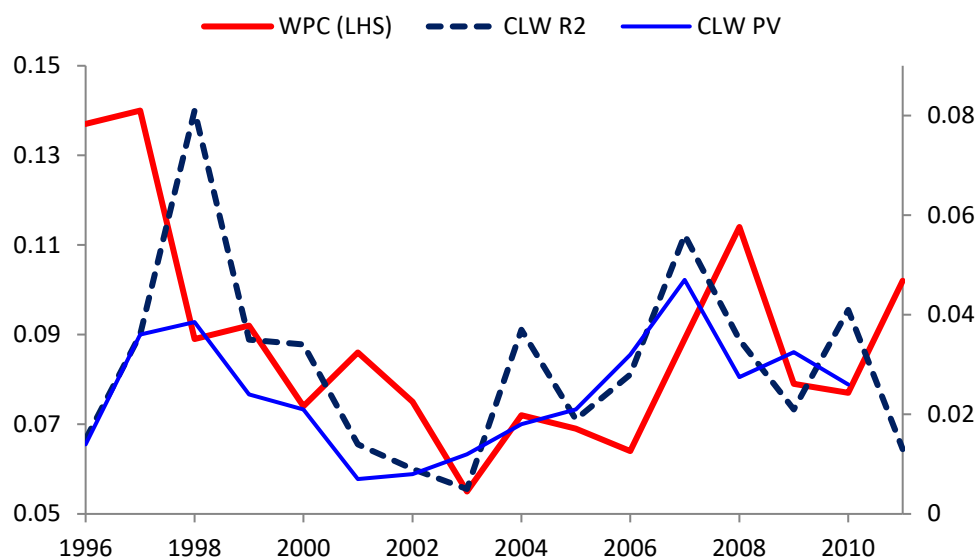


Figure 6: Annual Estimates of Price Delay

This figure depicts the annual estimates of price delay measures PD1, PD2, and PD3.

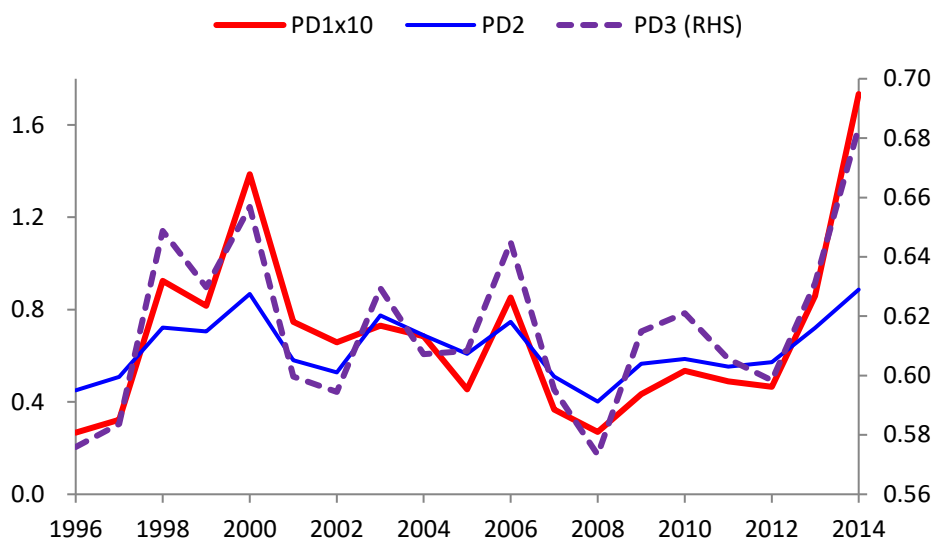
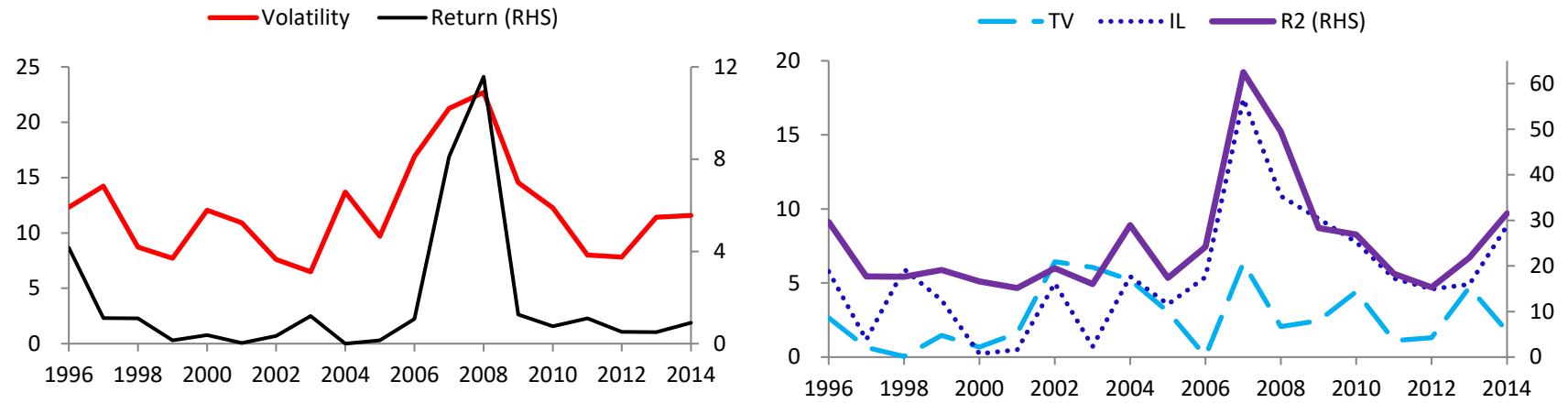


Figure 7: Partial R^2 of Market Condition Variables (%)

Panel A: WPC



Panel B: PD1

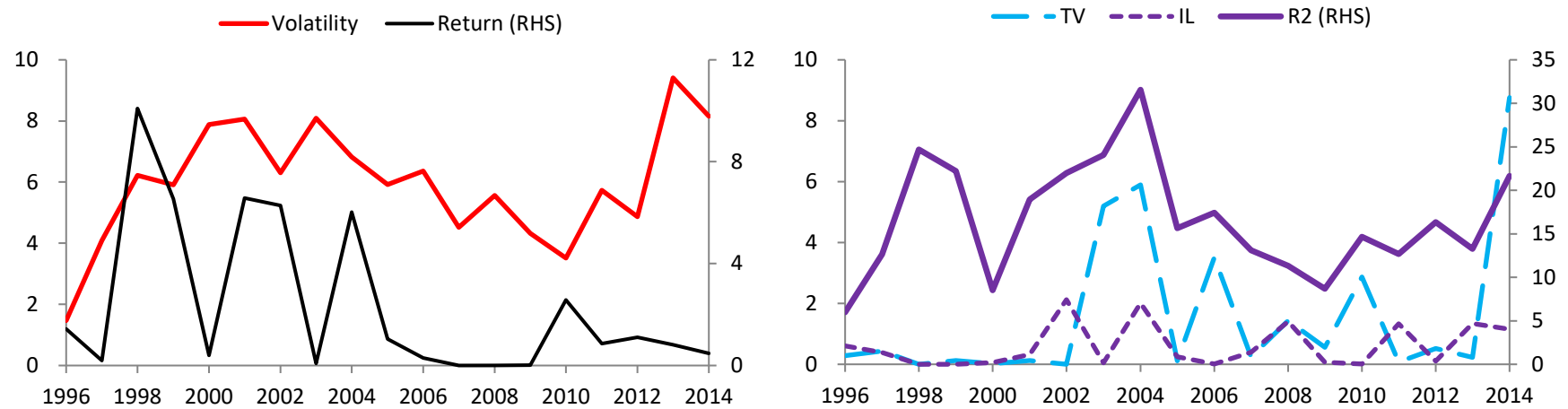
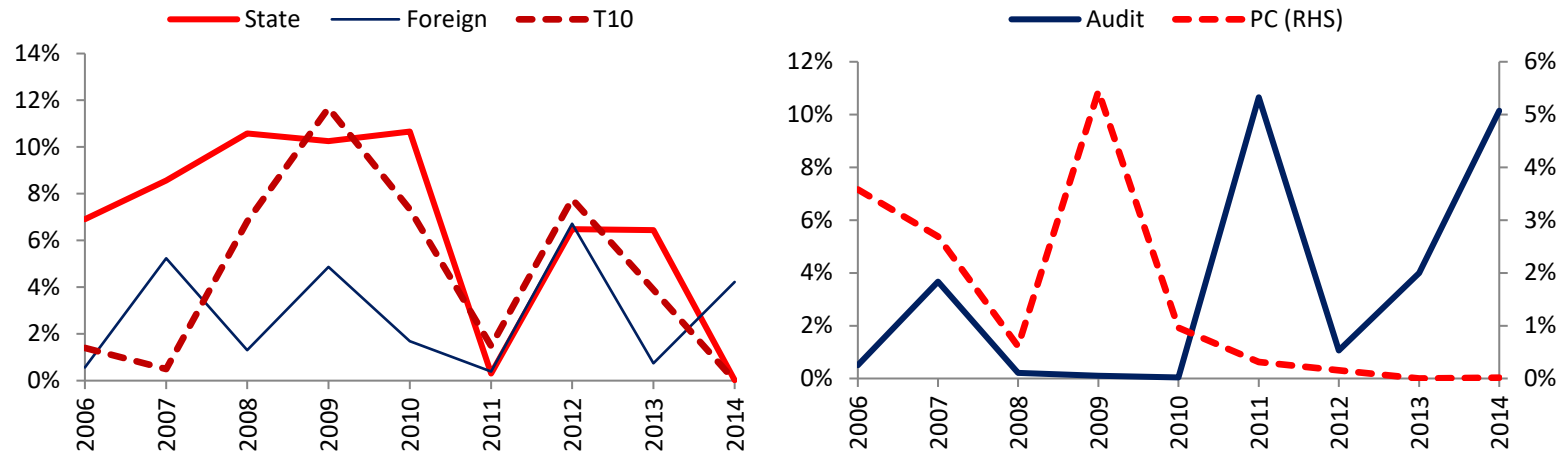


Figure 8: Partial R² of Corporate Governance Variables

Panel A: WPC



Panel B: PD1

